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Machine Learning Models Reveal Key Performance Metrics of Football Players to Win Matches in Qatar Stars League

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ABSTRACT As football (soccer) is one of the most popular sports worldwide, winning football matches is becoming an essential aspect of football clubs. In this study, we analyzed football players' performance in a total of 864 football matches of the Qatar Stars League (QSL) between the years 2012 and 2019. For each match, the collective performance of the players in key playing positions was analyzed to understand their effectiveness in winning games. We formulated this study as a classification framework in the machine learning (ML) context to distinguish the winning team from the losing team in a match. This allowed us to check the effectiveness of different performance metrics considered a feature vector for ML models. Different ML models were considered for this classification task, and the logistic regression-based model was considered the best performing model, with more than 80% accuracy. Multiple feature selection methods were leveraged to identify players' performance metrics that could be considered as contributing factors to determine the match result. The proposed ML model identified several features, including (a) shots on target by forwarders (b) distance covered by forwarders and midfielders at very high speed (c) successful passes, that can be considered as effective performance metrics for winning a football match in QSL. Interestingly, we revealed that the defenders' role could not be ignored for match results, and playing fair games improves the chance of winning matches in QSL. We also showed that players' performance metrics from the last five seasons would provide sufficient discriminative power to the proposed ML model to predict the match-winner in the upcoming season. The proposed ML model will support the players, coaching staff, and team management to focus on specific performance metrics that may lead to winning a match in QSL.

INDEX TERMS Football, soccer, players performance, machine learning, match result prediction, Qatar, Qatar stars league (QSL), Qatar football association (QFA), International Federation of Association Football (FIFA) World Cup.

I. INTRODUCTION

Football, also known as soccer, is the world's most popular sport [1]. The International Federation of Association Football (FIFA) estimated that football is played officially over 200 countries, and 1.3 billion football fans are supporting their teams globally [1]. Considering the financial perspective, the European football market alone is projected to exceed EUR 28 billion [2], and the football team management is continuously focusing on the selection of proper strategies to win matches in different leagues across the world. Each professional football team usually employ a group of analysts to measure the performance of their own players as well as opponent players [3]. There are many studies that have been conducted around the world focusing on several aspects of football including determining the factors to win a match. But the prediction of winner from a match is a daunting task. Some studies focused on playing tactics [4], sports medicine

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and care to avoid players injuries [5], and others investigated on players' physical and technical performance to win a match [6]. Recently many modern technologies have been introduced into football game to improve the quality of the matches such as using tracking wearable devices by players during official matches [7], the use of multi-camera tracking technologies, and the use of video assistance referee (VAR) system. Such technologies allow for the expansion of the match-related data collection, which can then, subsequently, be used for understanding the players' performance and analyze the match result in data-driven fashion. This data-driven approach would be tremendously useful to identify players' performance metrics, that are key to win matches in challenging football leagues. The data-driven approach can also help the team management to determine the potential game result (win or loss).

Most of the studies that highlighted on players' performance to win a match have been focused on competitions from top football leagues such as the "The Union of European Football Associations (UEFA) Champions League", European national leagues and the FIFA World Cup [8]. Very few studies have been focusing on other leagues around the world. In Qatar, there are some research studies that have been conducted focusing on preventive measures to avoid injury for the players in Qatar Stars League (QSL) [9]. Other studies focused on the effect of playing conditions (e.g., humidity, temperature etc.) on football players' injuries [10]. There are also some studies that have been focusing on players performance analysis in QSL [11]. However, no studies exist analyzing the football matches in QSL in data-driven fashion based on machine learning (ML) based techniques to understand the players' performance that may help a team to win a match in QSL.

The main objective of this study is to build machine learning models that can, potentially, be used to predict the winner from a football match in QSL. As part of this study, we came up with ML models that can achieve higher accuracy comparing to the other studies for same purpose. We also concentrated on identifying the best performance metrics that may lead to winning a football match in QSL. To the best of our knowledge, this is the first study focusing on players' performance to predict the winning team from a professional football match based on ML techniques in the Middle East and North Africa (MENA) region football leagues. We also suggested the performance metrics that lead to win the football match for the teams. We believe our study will help enhancing the performance of the players as well as facilitate the coaching staffs and team management in Qatar to make strategic plans which may lead to winning a football match. It will also help the team management in other football leagues across the globe to focus on specific performance metrics to enhance the chance of winning in football matches.

II. BACKGROUND STUDIES

Qatar is hosting the FIFA World Cup in 2022, which is the top football competition in the world going to be held for the very first time in the Middle East region. As a result, Qatar is investing heavily in QSL, which is the top ranked-football competition in the Middle East and the second best football competition in Asia [12]. Moreover, Qatar is emphasizing in skill developments for multiple aspects of football disciplines such as developing top-ranked referees worldwide who are hired in top-ranked competitions [13], having elite football development programs, and developing football academy like Aspire Academy which is approved as an elite academy by the Asian Football Confederation (AFC) [14]. Additionally, Qatar has started investing in research and development in the sports. But, unfortunately, the research works related to football in Qatar, as well as in MENA region, is still in its infancy.

There are many existing methods to capture the metrics related to players' performance in a football match from European leagues. Some methods use match video, which could be achieved by media channels, having 24 up to 70 cameras in a stadium from multiple angles, to extract data for players' performance [15]. Companies like Stats Perform [16], and Opta [17] use their own recordings with advanced image and video processing system to collect players' performance data in a match. For this purpose, they use a multi-camera system installed in stadiums to track the player during official matches [18]. These systems are highly accurate in capturing the performance metrics for the analysis of football players' movement patterns in the football pitch [15], [19]. Some other frameworks use wearable sensors attached directly to players, and match equipment (e.g., ball) to collect data during match time [15]. In QSL, Stats Perform is used to capture the performance metrics for match time players' performance metrics.

There are several studies where ML based techniques have been used to predict the winner from a football match. Yezus used four ML models (k-Nearest Neighbors (kNN), Random Forest (RF), Logistic Regression (LR), and Support Vector Machine (SVM)) to predict the outcome of a football match on the English Premier League [20]. The study used open data from "championat.com" and "statto.com" to extract features for ML models. The study used nine features and 640 records to obtain 63% accuracy for the best performing RF based model. Prasetio *et al.* proposed an LR based model to predict the winner of the English Premier League [3]. The study used match data from open public websites "football-data.co.uk" and "sofifa.com/teams" for several seasons (2010/2011 to 2015/2016). The performance of the model was evaluated on the latest season, while the model was trained using data from the rest of the seasons. The proposed model achieved an average accuracy of 69.5%. Martins *et al.* used ML models to predict the outcome of football matches using dataset from the English Premier League (2014-2015), the Spanish La Liga (2014-2015), and the Brazilian League Championships (2010-2012) [21]. Different features were extracted for each league and then consolidated into one dataset. Then, polynomial classifier has been used to expand the input features into high dimension. The authors applied different ML methods

(naïve Bayes (NB), Decision Trees (DT), Multilayer Perception (MLP), Radial Basis Function (RBF), and SVM) for predicting the outcome. The proposed MLP based model achieved 100% accuracy. It is important to stress that the authors used ''goal'' as one of the features in building ML models, which helps to achieve a high level of accuracy in predicting the match result. Zaveri *et al.* used different ML models (LR, RF, Artificial Neural Network (ANN), Linear SVM, and NB) to predict match results from the Spanish La Liga [22]. The authors used 13 different features (e.g., shots on target, corners, yellow cards, etc.) based on home and away games to predict match results from season 2012 to 2017 and the best performing model achieved 71.6% accuracy. Tüfekci [23] used 70 different features to build ML models (SVM, bagging with REP Tree (BREP), and RF) to predict three outcomes: (1) Home Win (2) Draw (3) Away Win, on the matches from the Turkish Super League. The study used publicly available dataset from ''mackolik.com'', ''fenersk.com'', and ''ftt.org'' covering four seasons starting from 2009 up to 2013 with a total of 1,222 matches. The author categorized the features based on home team playing at home conditions, home team is playing away, away team playing at home conditions, and away team playing at away conditions. For each of these categories, several features have been used such as number of games played, games won, total goals, and goals per game. RF based model was the best performing model with 71% accuracy. Mao *et al.* focused on the Chinese Super League from the 2014-2015 season which included 480 matches [8]. Twenty-one performance-related variables were extracted for the study from publicly accessed websites such as ''sina.com'', ''sohu.com'' and others. Then the collected information was categorized into four groups: (a) goal scoring variables (b) passing and organizing variables (c) defending variables (d) contextual variables. The authors used cumulative approach to predict the result of a match. The authors concluded that five variables had a consistent effect in the different match contexts. Two of these variables,''Shot on Target'' and ''Shot Accuracy'', had a positive effect on the performance. The other three variables ''Cross Accuracy'', ''Tackle'', and ''Yellow Card'' were trivial. All the other 16 variables varied depending on the strength of the team and its opponent. Zhange *et al.* [24] considered 1,440 matches from the Chinese Super League starting from season 2013 up to 2016. They used fourteen features and three different ML models (SVM, RF, and Bayesian network) to predict the match result with 76.9% accuracy.

Deep Learning (DL) based approach has been used by Pettersson and Nyquist [25] to predict the outcome of a football match. The authors considered the dataset of football leagues and tournaments from 54 different countries. It included more than 35,000 matches from 2015 to 2017. The authors used long short-term memory (LSTM) model as the proposed ML model to predict the outcome of the matches. The study used seven different models, where each model used a different number of layers and LSTM units. For each model, authors were predicting the match outcome for every 15 minutes interval of the match duration (90 minutes). The final prediction accuracy for the models ranged from 77.8% to 98.6%. It is important to stress that the model used the number of goals as one of the features in building ML models, which helped to achieve a high level of accuracy in predicting the match result. Another DL based study was done in 2020 [26] to predict the outcome of the FIFA World Cup 2018 matches using publicly available datasets ''International Football results from 1872 to 2018'', ''FIFA Soccer Ranking'', and ''FIFA World Cup 2018 Dataset'' and ''FIFA Soccer Ranking'' dataset. The authors proposed ANN and LSTM based model to predict the outcome of the matches with 63.3% accuracy. Danisik *et al.* used LSTM to predict match outcomes [27]. The authors built a dataset from multiple sources and categorized their features in terms of player statistics and match history. The dataset included matches from different European Leagues from 2011 to 2016. They achieved an accuracy of 52.4%.

A summary of the studies, focusing on determining the winning team from a football match, is highlighted in Table 1. In Table 2, we listed the different ML models used in each study that was discussed in this section.

III. MATERIALS AND METHODS

A. ETHICAL APPROVAL

This study was conducted under the regulation of the Ministry of Public Health, Qatar. All procedures were approved by the Institutional Review Board (IRB) of Qatar Biomedical Research Institute (QBRI), Hamad Bin Khalifa University (HBKU), Qatar and only de-identified data were collected from QSL.

B. DATA COLLECTION FROM QATAR STARS LEAGUE (QSL)

We collected data for the professional football matches of the QSL. QSL is the top-ranked league in Qatar, which involves a range of 12 to 14 teams per season. The season usually starts from August/September and ends in February/March. Each team in the season plays against all the other teams in the league and plays two matches with each of the other teams in a home and away style. The winner of the competition is determined by calculating the number of points a team has where they get three points for a win, one point for a draw, and no points when losing. For the QSL match time, players' performance metrics were captured using Stats Perform [16] platform installed in multiple stadiums around Qatar. The data considered for this study were collected between the years 2012 to 2019 for the seven consecutive football seasons: (a) 12/13 (b) 13/14 (c) 14/15 (d) 15/16 (e) 16/17 (f) 17/18 and (g) 18/19. The dataset contains a record of 2,419 players who played in 18 teams in a total of 1,121 matches (Table 3).

In the dataset, there were 34 different measurements [16], which we split into four main categories (Table 4). The first category contained the player ID, his team, opponent, and the playing position in a match. The second category covered match-related information, and it contained match

TABLE 1. Summary of ML studies to predict the outcome of football matches.

Study	Year of Study	Competitions	Years	Accuracy	Important Variables	Brief Summary
[20]	2014	English Premier League	NA	63%	NA	Used different ML methods to predict the outcome of matches using special formula by the authors
[23]	2016	Turkish Super League	2009-2013	70.9%	General match information such as Derby Game, Penalty without Spectators, and Stadium Ban	The authors used feature selection techniques along with different ML models to predict the outcome of football matches
[8]	2016	Chinese Super League	2014-2015	NA	Shot on Target and Shot Accuracy	The study focused on identifying the features that lead to win a match
[3]	2016	English Premier League	2010-2016	69.5% Home Defense and Away Defense	LR model was used to predict the outcome of the matches	
[25]	2017	Leagues and Competitions from around the world	2015-2017	100%#	NA	Prediction of result based on goals# and real-time match events at 15 minutes interval. The likelihood of the correct prediction improves gradually over the match period
[21]	2017	English Premier League, La Liga, Brazilian League Championships	2010-2015	99%#	NA	Prediction of result based on players' performance and goal# in match
[27]	2018	Different Leagues in Europe	2011-2016	52.4%	NA	The authors used LSTM to predict the outcome of a football match
[22]	2018	Spanish La Liga	2012-2017	71.6%	Team vs. Away Team Win percentage	The authors used different ML models to predict the outcome of a football match
[24]	2019	Chinese Super League	2013-2018	76.9%	NA	The authors used different ML models to reach their best accuracy
[26]	2020	Different Leagues	1872-2018	63.3%	NA	The authors wanted to predict the outcome of FIFA World Cup 2018 using different ML models

NA = Not Available (information not specified)
#This model used "goal" as a feature to develop machine learning models

TABLE 2. ML models used in previous studies.

Machine Learning Model	Reference
k-Nearest Neighbors (kNN)	[20]
Random Forest (RF)	[20], [22]–[24]
Logistic Regression (LR)	[3], [20], [22]
Support Vector Machine (SVM)	[20]–[24]
Bagging with REP Tree (BREP)	[23]
Long short-term memory (LSTM)	[25]–[27]
Naïve Bayes (NB)	[21], [22]
Decision Trees (DT)	[21]
Radial Basis Function (RBF)	[21]
Bayesian	[24]
Multilayer Perception (MLP)	[21], [22], [26]

TABLE 3. Dataset summary.

Season	Total Matches	Win-Lose Matches	Number of Teams	Number of Players
12/13	132	102	12	321
13/14	182	130	14	351
14/15	181	140	14	350
15/16	182	143	14	364
16/17	181	135	14	374
17/18	132	105	12	327
18/19	131	109	12	332

result, location, season, etc. The third category (''technical performance'') contained the technical performance metrics (e.g., shots, fouls, dribbles, etc.) for each player in a match. Finally, we categorized six metrics into ''physical performance'' group representing players' physical activity in a match.

In our dataset, the ranking of the teams were not available. Also, for the substitute players, we were missing almost all of the features of interest including the position of the substitute players. As we were interested in position specific analysis, we discarded the information of substitute players from our study.

C. DATA CLEANING AND PRE-PROCESSING

From the match related metrics, we discarded the following measurements as they were not useful for this study:

- Competition: The name of the competition.
- Round: The round when the match was conducted.

TABLE 4. List of measurements available in the dataset.

Measurement	Description
Player Data	
Player ID	The ID of a player
Team	The team which the player is playing in
Opponent	The opponent team in the match
Position	The position of player based on the coach plan
Match Data	
Competition	The competition which the match is part of
Season	The season of the match
Round	The round within a season that the match is part of
Match Date	The date when the match was played
Venue	The venue or stadium where the match is played
Match	The match details where the home and away team is specified in the format (Team 1 vs Team 2)
Scoreline	The final score of match in the format (x : y) where x and y are numerical numbers
Result	The result of the match indicating if the team of the player wins, losses, or draws
Technical Performance	
Blocks	The number of times a player attempts ball blocks
Clearances	"An attempt to move the ball away from the current area, keeping possession is lower priority" [16]
Corners	The number or corners kicks the player did
Crosses	"A pass to a teammate that originates from the offensive flank aimed at the area in front of the goal" [16]
Dribbles	"Times the offensive player while advancing the ball attempted to maneuver around a defender in close proximity" [16]
Fouls	"When a player commits a foul for kicking, tripping, jumping in/at, charging, striking or pushing an opponent" [16]
Free Kicks	"Number of indirect and direct free kicks" [16]
Interception	"Shots, Controls, Passes and Clears that were marked as Interceptions" [16]
Offsides	"Number of times player/team was caught offsides" [16]
Red Cards	The number of times a player got a red card
Shots	"An attempt to score a goal with any part of an attacking player's body" [16]
Shots on Target	An attempt to score a goal with any part of a player's body that would go in the goal if nobody saved, blocked, or deflected it [16]
Passes	"An attempt to play the ball to a teammate" [16]
Successful Passes	"Number of passes that are followed by a contact with a teammate" [16]
Tackles	"A fully committed attempt to dispossess an opponent while the ball is under hip height" [16]
Yellow Cards	The number of times a player got a yellow card
Physical Performance	
Standing Distance	Distance covered by player in the speed of 0-1 km/hr [16]
Low Speed Distance	Distance covered by player in the speed of 1-6 km/hr [16]
Moderate Speed Distance	Distance covered by player in the speed of 6-15 km/hr [16]
Elevated Speed Distance	Distance covered by player in the speed of 15-20 km/hr [16]
High Speed Distance	Distance covered by player in the speed of 20-25 km/hr [16]
Very High Speed Distance	Distance covered by player in the speed of 25^{+} km/hr [16]

All the measurements under Technical and Physical Performance are numerical

- Venue: Venue of the match.
- Opponent: The name of the opponent team.
- Scoreline: The actual score of the winning and losing team is not relevant. Only the final result, which is win, lose, draw is considered.
- Match date: The date when the match happened.

To model position-specific metrics, we categorized the positions of a player during the match into four categories. These categories are: (a) forwarder (b) midfielder (c) defender (d) goalkeeper. For some players, positions data were missing in the dataset, and we discarded such records from the analysis. We also discarded records for the player with the goalkeeper as position as we think that the goalkeeper performance has different metrics than the other players due to the different activities that they have during the match. We converted the match result into a numerical value using the following rule: (a) Win = 1 (b) Loss = 0 (c) Draw = −1. Draw matches were excluded as this is out of the scope of this study. So, if Team_A wins a game against Team_B, then Team_A will be considered as a positive class sample for that particular match and the Team_B will be considered as a negative class sample for the same match.

D. FEATURE ENGINEERING FOR THE PLAYERS' PERFORMANCE METRIC IN MATCHES

First, we grouped the players based on three different playing positions (forward, midfield and defense) in a match. We considered the sum of the 16 metrics (blocks, clearances, corners, crosses, dribbles, fouls, free kicks, interceptions, offsides, red cards, shots missed, shots on target, successful passes, tackles, unsuccessful passes, and yellow cards) to generate 16 aggregated metrics for all players playing in a match. Based on three different playing positions (forward, midfield, or defense), we generated 48 (= 3*16) different features.

Then, we calculated the mean value for six different metrics (elevated speed, high speed distance, low speed distance, moderate speed distance, standing distance, and very high-speed distance) to generate six aggregated metrics for all players playing in a match. Based on three different playing positions (forward, midfield, or defense), we generated 18 (= 3*6) features.

Additionally, we added the total number of players playing at three different positions (forward, midfield, or defense) in a match generating three more features.

We considered the effect of performance from the opponent team and checked if that would contribute in determining the winner from a football match as well. For this purpose, we converted the aggregated performance metrics of a team based on the performance of the opponent team by taking the percentage of the performance metrics for both teams during the match. For example, in a match, if the total shots on target by forwarders from winning team was shot_W, and the total shots on target by forwarders from losing team was shot_L, then we generate a feature holding a value of $(\text{shot}_W)/(\text{shot}_W+\text{shot}_L)$ for the winning team and $(\text{shot}_L)/(\text{shot}_W+\text{shot}_L)$ for the losing team. In this way, the corresponding feature values were representing the proportion of total shots on target by forwarders from both teams in a match. The final dataset included 69 features and the class label, as detailed in the list in Supplementary Table 1.

E. FEATURES NORMALIZATION

The features were normalized using min-max normalization [28] as in 1.

$$x' = \frac{x - min(x)}{max(x) - min(x)} \tag{1}$$

where x is the value of a feature, $min(x)$ is the minimum value of feature x, and $max(x)$ is the maximum value for the feature x.

F. STATISTICAL ANALYSIS

As part of exploratory data analysis, normal distribution hypothesis testing was conducted on all the features. First, "Anderson-Darling" (AD) test [29] was done on all features to check if the features are normally distributed. We found out that five features were normally distributed. For the normally distributed features, we applied a "Student T-Test" [30] to check if there is a statistically significant difference for such features between the winning team and the losing team. For the same purpose, we applied "Mann-Whiteny"(MW) [30] test on the reaming 64 features that were not following normal distribution. Detailed analysis of each of the features along with the corresponding p-value for each feature is highlighted in Supplementary Table 1. Out of these 69 features, 47 features were statistically significant when comparing the winning team vs. losing team, and 22 features were not.

G. MODELLING

We developed several machine learning classification models to determine the winning (class = 1) and losing team (class = 0) from a match. To develop such classifier, we used nine different machine learning algorithms from the "scikit-learn" library in Python: kNN [31], RF [32], [33], DT [33], LR [3], MLP [34], SVM with linear kernel (L-SVM) [34], [35], SVM with polynomial kernel (P-SVM) [35], SVM with radial basis function kernel (RBF-SVM) [35], and XGBoost [36].

H. MODEL EVALUATION

We used five-fold cross validation by splitting the 80% data as training dataset and 20% as validation dataset. We used the following performance evaluation metrics: (i) Accuracy as in (2), (ii) Sensitivity as in (3) (iii) Specificity as in (4) and (iv) Area Under the Curve (AUC) from the Receiver Operating Characteristics (ROC) curve to analyze the performance of different machine learning models.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{3}$$

$$Specificity = \frac{TN}{FP + TN} \tag{4}$$

where TP, FN, FP, and TN stand for true positive, false negative, false positive, and true negative, respectively

I. MODEL PARAMETER OPTIMIZATION

To apply hyper-parameter tuning for all the models, we applied "Grid Search" with cross validation method [37] on all the classifiers. We set the following parameters for the machine learning classifiers:

- kNN: "leaf_size" = 1, "n_neighbors" = 47, "p": 1, "weights": "uniform"
- RF: "max_depth" = 31, "max_leaf_nodes" = 17, "n_estimators" = 36, "criterion": "gini", "max_features": "sqrt"
- DT: "criterion" = gini, "splitter" = "best"
- LR: "C" = 5.0, "solver": "saga", "tol": 0.001, 'class_weight': 'balanced', 'multi_class': 'ovr', 'penalty': 'none', 'warm_start': 1
- MLP: "alpha" = 0.001, "activation": "tanh", "hidden_layer_sizes" = 100
- L-SVM: "C" = 4.5, "degree" = 0, "gamma" = 1e-05
- P-SVM: "C" = 2.7, "degree" = 1, "gamma" = 0.2
- RBF-SVM: "C" = 1.3, "degree" = 0, "gamma" = "auto" (1 / n_features)
- XGBoost: "gamma" = 1.0, "max_depth" = 3, "min_child_weight" = 1, "subsample" = 0.9

J. FEATURE SELECTION

Feature ranking is an essential step for any classification task where the goal is to select only those features that are rich in discriminatory power with respect to the classification problem at hand [38]. The idea of feature ranking is to exploit the information included in the dataset (e.g., the correlation between variables and discriminatory abilities of the individual features) to consider the most promising feature subset by discarding the irrelevant ones for the task of learning. In order to rank the features, we applied "Pearson Correlation Coefficient" (PCC) [39] method on all the features comparing each feature to the "Result" (class label) which was calculated using the equation in 5.

$$\rho_{a,b} = \frac{E(a, b)}{\sigma_a \sigma_b} \tag{5}$$

where:

- E is the covariance
- σ_a is the standard deviation of a
- σ_b is the standard deviation of b

We also applied decision tree-based Gini Index (GI) measurements, Random Forest based Index (RFI) and coefficients of LR model to rank the features. Supplementary Table 1 highlights the full list of features and their ranking based on different ranking methods. We also applied the recursive feature elimination (RFE) technique [40] to select a subset of features from the pool of all features (N = 69).

IV. RESULTS AND DISCUSSION

A. PERFORMANCE OF THE ML MODELS

For the analysis, we have combined the matches from all seasons and split them into training and test set randomly without

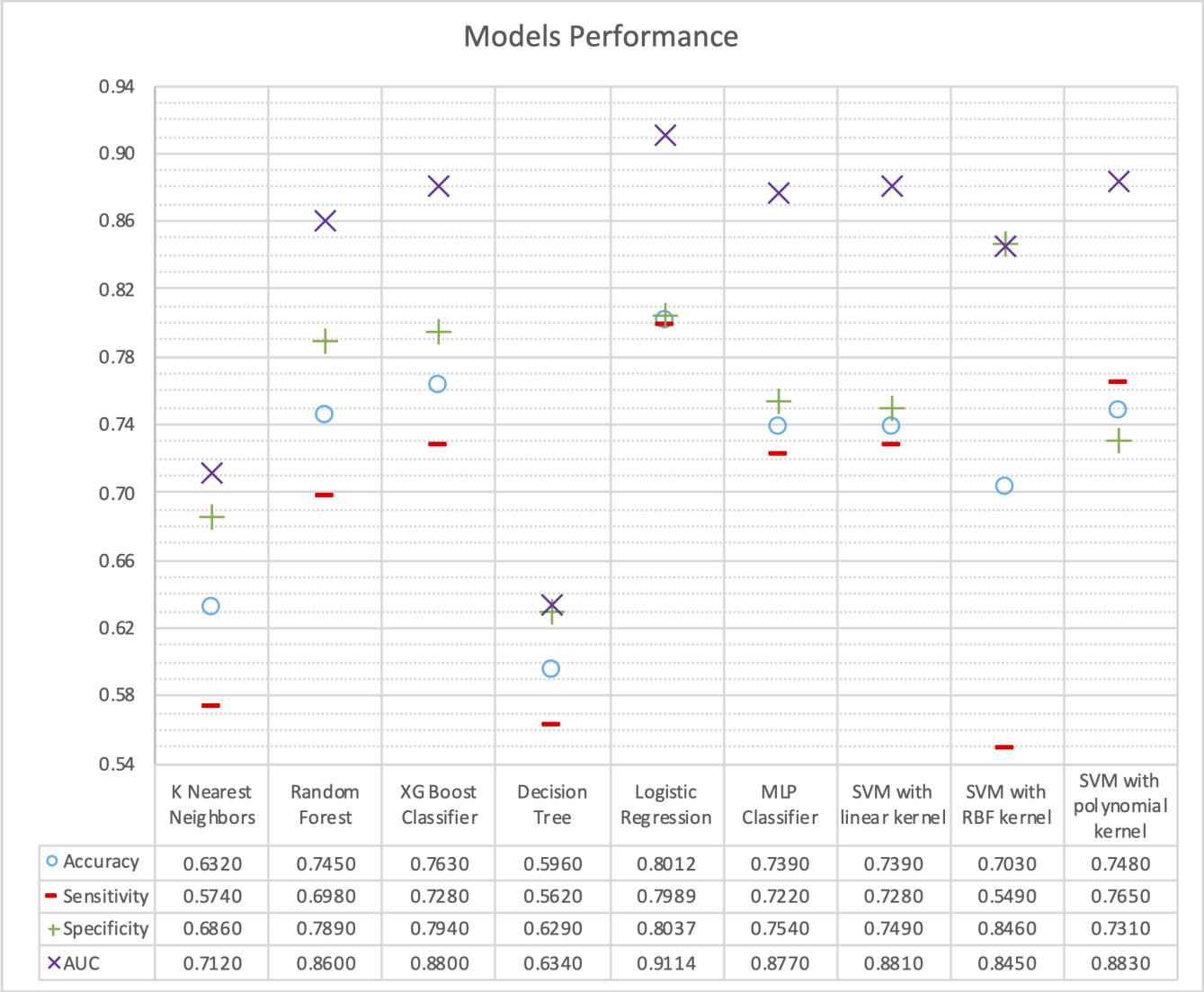

FIGURE 1. **Performance of different machine learning models for distinguishing winning from the losing team.**

considering any season specific information. We summarized the results that we achieved using nine different models on the dataset in Fig. 1. It shows the accuracy, sensitivity, specificity, and area under the curve (AUC) for each model graphically. From Figure 1, we can observe that the LR based model was the best performing model in classifying the winning team from the losing team. The model achieved over 79.9% sensitivity, 80.4% specificity, and 80.1% accuracy. The highest AUC over 0.91 was obtained based on the LR based model as well.

SVM with linear kernel and SVM with polynomial kernel models performed at a similar level. This SVM with linear kernel achieved over 72% sensitivity, 74% specificity with 73% accuracy. This SVM with polynomial kernel achieved over 76% sensitivity, 73% specificity with 74% accuracy. The SVM with the RBF kernel model achieve the highest specificity of 84%, but the sensitivity was on a lower side of only 54%, and accuracy was over 70%.

Ensemble-based models RF and XG Boost classifiers showed similar performance. RF model achieved over 69% sensitivity, 78% specificity, and 74% accuracy. XG Boost classifier performed slightly better than RF with over 72% sensitivity, 79% specificity, and 76% accuracy.

Out of all the classifiers we investigated, the proposed LR based classifier achieved the best accuracy in distinguishing match winner form the loser. LR assumes the logit transformation of the output variable (class label) has a linear relationship with the independent variables (feature vector). LR also requires little or no multicollinearity among the independent variables (feature vector), and the prediction performance of an LR classifier depends upon whether the underlying dataset holds the assumed model [41]. As the proposed LR-based classifier performed at a higher level of

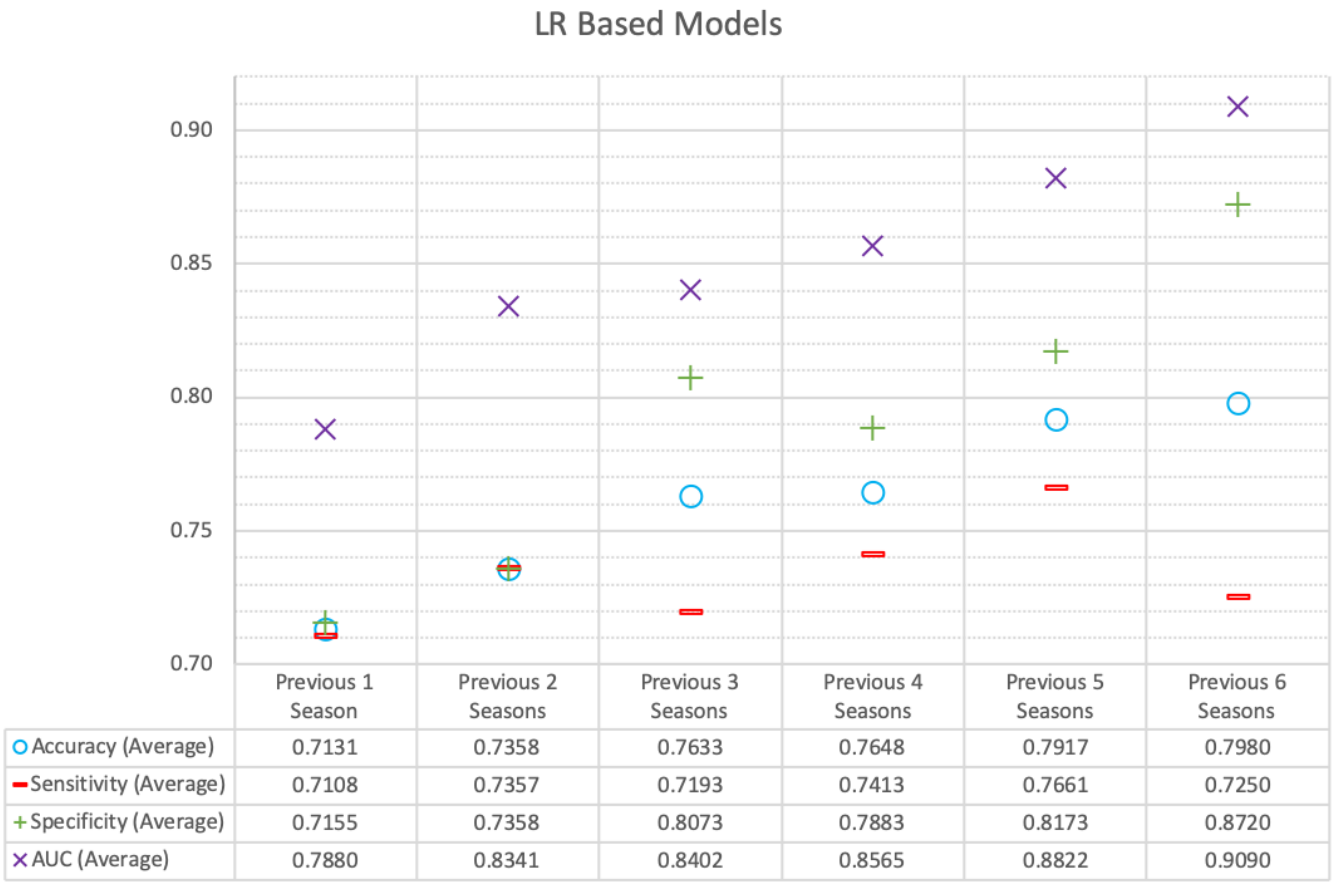

FIGURE 2. Performance of LR based model to predict the match result for upcoming season.

accuracy than the other ML-based classifier, we think the dataset from QSL represented the desirable properties of the LR classifier. But we cannot generalize the superiority of the LR classifier for all other datasets as the proposed classifier was developed based on a specific dataset from QSL.

We applied RFE to select a subset of features from within a range of two features up to all features (N = 69). And we got the best accuracy of 79.39% using 31 features for LR (Supplementary Table 2). Though the model with a smaller number (N = 31) of features performed at an accuracy level similar to the model with all features, it could not outperform the model with all features (N = 69). So, we considered the model with all features as the best performing model. Supplementary Figure 1, shows the ROC curves for all the models we evaluated for this study.

B. KEY PERFORMANCE METRICS THAT CONTRIBUTED IN WINNING MATCHES IN QSL

Based on the feature ranking methods we adopted, we found shots on target by forwarder was the most important feature determined by all the three methods (PCC, GI, and RFI) (Supplementary Table 1). This finding is conceivable as the number of shots on target by forwarders increases, the chance of scoring a goal improves as well. This will ultimately help a team to win a match. In our study, we found that the shots of target by the forwarders from winning team (W) was higher than the same from the losing side (L) (W:L = 0.60:0.38, p-value = 1.02e−54).

All four feature ranking methods (PCC, GI, RFI, and LR) considered successful passes made by players playing at three positions (defense, midfield, forward) as key features for the classification framework. We observed that the number of successful passes made by winning team players were higher than the players from the losing team (for forward, W:L = 0.52:0.47, p-value = 3.56e−11; for midfield, W:L = 0.52:0.47, p-value = 1.34e−09; for defense, W:L = 0.53:0.46, p-value = 4.81e−14).

All four feature ranking methods (PCC, GI, RFI, and LR) considered the distance covered by the players playing at high speed as key features for the classification framework. Winning team players from all three positions (defense, midfield, forward) covered more distance at very high speed (25^{+} km/hr) compared to the losing team, and the difference was statistically significant. The distance covered by forwarders at very high speed from the winning team was higher than the losing side (W:L = 0.51:0.45, p-value = 6.23e−19). Similarly, distance covered by midfielders at very high speed from the winning team was higher than the losing

team (W:L = 0.50:0.46, p-value = 4.46e−11). This indicates the contribution of very high speed from forwarders and midfielders in winning football matches in QSL.

We also observed that defenders played a crucial role in winning matches in QSL. Shots on target by defenders, clearance made by defenders, and successful passes made by the defenders were among the top-ranked ten features based on PCC, GI, and LR. Clearance made by the defenders (W:L = 0.52:0.47, p-value = 6.36e−10) and the successful passes made by the defenders (W:L = 0.53:0.46, p-value = 4.81e−14) were higher for the winning team compared to the losing team. Interestingly, shots on target by defender were highly correlated to the result of a match (PCC = 0.23), and it was significantly higher for the winning team compared to the losing side (W:L = 0.54:0.36, p-value = 2.65e−22). Therefore, we conclude that not only forwarders but defenders also play a crucial role in winning matches in QSL.

Our analysis revealed that teams played fair games in football matches had a higher chance of winning in QSL. This conclusion was based on our observation that defender and midfielders from the winning teams faced a lower number of yellow cards compared to the losing team (for defender, W:L = 0.41:0.45, p-value = 0.010; for midfielder, W:L = 0.27:0.34, p-value = 0.0002). Similar trend was observed for red card as well (for defender, W:L = 0.03:0.07, p-value = 3.39e−06; for midfielder, W:L = 0.01:0.03, p-value = 0.016).

C. PERFORMANCE OF THE MODEL FOR PREDICTING MATCHES IN UPCOMING SEASON

We trained the models using data from the last two to six seasons and evaluated the model's performance in predicting match results for the upcoming season. We found that after training models based on last one-, two-, three-, four-, five-, six-years data, the LR model was able to predict the match result at over 71%, 73%, 76%, 76%, 79% and nearly 80% average accuracy for the upcoming season (Table 5, Fig. 2). So, there was almost a monotonic trend of performance (accuracy) improvement by incorporating more seasons into the training set.

The model's performance was very close when we considered the previous five (accuracy: 79.2%) or six seasons (accuracy: 79.8%) as training dataset. Compared to the five-season based model, the six-season based model achieved an improvement of specificity (∼ 5%) at the cost of sensitivity (∼ 4%). And considering the actual number of misclassified instances (matches), the drop of sensitivity by 4% (representing nearly four misclassified matches from positive class) does not affect the six-season based model's overall performance as we considered only 109 games from season 18/19 as testing dataset (Fig. 2). As the accuracies from these two experimental setups (five- and six-season based models) were very close, we concluded that information from the previous five seasons would be good enough to train a model to predict the match result for the upcoming season.

TABLE 5. LR based model to predict the match outcome for upcoming season.

Seasons in Training set	Seasons in Testing set	Accuracy	Sensitivity	Specificity	AUC
Previous 1 season					
17/18	18/19	0.757	0.78	0.734	0.842
16/17	17/18	0.757	0.762	0.752	0.815
15/16	16/17	0.659	0.748	0.57	0.74
14/15	15/16	0.673	0.535	0.81	0.725
13/14	14/15	0.745	0.763	0.727	0.836
12/13	13/14	0.688	0.677	0.7	0.77
	Average	0.713	0.711	0.716	0.788
Previous 2 seasons					
16/17 – 17/18	18/19	0.716	0.936	0.495	0.867
15/16 – 16/17	17/18	0.757	0.743	0.771	0.848
14/15 – 15/16	16/17	0.781	0.815	0.748	0.835
13/14 – 14/15	15/16	0.658	0.423	0.894	0.759
12/13 – 13/14	14/15	0.766	0.763	0.77	0.862
	Average	0.736	0.736	0.736	0.834
Previous 3 seasons					
15/16 – 17/18	18/19	0.817	0.853	0.78	0.891
14/15 – 16/17	17/18	0.776	0.733	0.819	0.868
13/14 – 15/16	16/17	0.77	0.756	0.785	0.822
12/13 – 14/15	15/16	0.69	0.535	0.845	0.779
	Average	0.763	0.719	0.807	0.84
Previous 4 seasons					
14/15 – 17/18	18/19	0.784	0.734	0.835	0.893
13/14 – 16/17	17/18	0.762	0.705	0.819	0.861
12/13 – 15/16	16/17	0.748	0.785	0.711	0.816
	Average	0.765	0.741	0.788	0.856
Previous 5 seasons					
13/14 – 17/18	18/19	0.812	0.78	0.844	0.904
12/13 – 16/17	17/18	0.771	0.752	0.79	0.861
	Average	0.792	0.766	0.817	0.882
Previous 6 seasons					
12/13 – 17/18	18/19	0.798	0.725	0.872	0.909

V. RESEARCH IMPLICATIONS

Our study revealed key performance metrics from the players in QSL which, ultimately, supported the teams to win football matches in QSL. Shots on target by forwarders was among the most important features and we recommend the forwarders to improve the accuracy of shots on target to enhance the chance of winning matches. Our study also emphasizes the contribution of defenders in terms of their clearance, successful passes and shots on target to win matches in QSL. This indicates that defenders are also crucial for not only clearing the ball, but they can also contribute in scoring to improve the chance of winning a match. We also recommend the teams to improve the fitness level of the players as distance covered by the players at high speed is very important in winning match. Additionally, accuracy of passes from all three positions (defense, midfield, forward) should be improved to win matches. Lastly, we recommend the teams to practice fair play policy by avoiding unnecessary fouls and injuries in the games. Playing fair games will not only empower the spirit of the games but also improve the chance of winning football matches in QSL. We believe our findings would support the football team management in Qatar to design long term strategic plan for the improvement of the match performance in club level and international level as well.

VI. LIMITATION

There are a few limitations to our study. Due to the unavailability of the similar data from other football leagues, we conducted our study based on performance metrics collected from QSL only. So, the performance of the model was not evaluated for other football leagues. The home/away factors for matches were not considered as we think it is not applicable in Qatar as the stadiums are relatively close to each other, and teams are not playing in their home or away stadiums consistently. Moreover, not all the teams from QSL own their stadium, and many matches are played at stadiums, which are not owned by any specific football club. So, the concept of home and away games is not fully applicable to QSL. There are some external factors like grass and ball types, that may affect the match results. But we ignored such factors as those external factors are standardized in QSL.

VII. CONCLUSION

In this study, we used ML-based techniques to predict the football match results in QSL using players' performance metrics. As per our best knowledge, this study is the first one in the MENA region to predict the winning team for a football match considering player's performance metrics from professional football league. The proposed model was able to reach up to 80% accuracy in distinguishing the winning team from the losing team for QSL football matches. Proposed ML models also suggested intriguing factors that may contribute to winning football matches in QSL. We showed that shots on target by the forwarder, distance covered by forwarder and midfielder at very high speed, successful passes by the players, and playing fair games might improve the chance of winning football matches in QSL. Finally, we proposed that developing ML model based on players' performance metrics from the previous five seasons would provide sufficient distinguishing capability to the model to predict the match outcome for the upcoming season. In the future, we will incorporate the drawn matches into the development of machine learning models to predict win, lose and drawn matches.

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